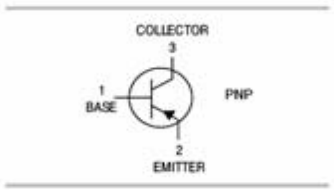
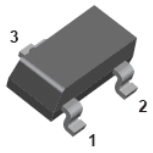
Silicon Epitaxial Planar Transistor **S8550**

FEATURES

- High Collector Current.(I_C= -500mA)
- Complementary To S8050.
- Excellent H_{FE} Linearity.

Lead-free





SOT-23

APPLICATIONS

- High Collector Current.

ORDERING INFORMATION

Type No.	Marking	Package Code
S8550	2TY	SOT-23

MAXIMUM RATING @ Ta=25℃ unless otherwise specified

Symbol	Parameter	Value	Units
V_{CBO}	Collector-Base Voltage	-40	V
V_{CEO}	Collector-Emitter Voltage	-25	V
V_{EBO}	Emitter-Base Voltage	-5	V
I_C	Collector Current -Continuous	-500	mA
P_C	Collector Dissipation	300	mW
T_j,T_{stg}	Junction and Storage Temperature	-55~150	℃

Silicon Epitaxial Planar Transistor **S8550**

ELECTRICAL CHARACTERISTICS @ Ta=25℃ unless otherwise specified

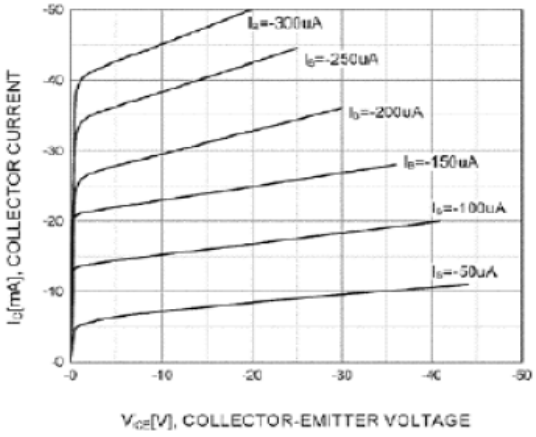
Parameter	Symbol	Test conditions	MIN	MAX	UNIT
Collector-base breakdown voltage	$V_{(BR)CBO}$	I_C=-100μA,I_E=0	-40		V
Collector-emitter breakdown voltage	$V_{(BR)CEO}$	I_C=-1mA,I_B=0	-25		V
Emitter-base breakdown voltage	$V_{(BR)EBO}$	I_E=-100μA,I_C=0	-5		V
Collector cut-off current	I_{CBO}	V_{CB}=-40V,I_E=0		-0.1	μA
Collector cut-off current	I_{CEO}	V_{CE}=-20V,I_B=0		-0.1	μA
Emitter cut-off current	I_{EBO}	V_{EB}=-3V,I_C=0		-0.1	μA
DC current gain	h_{FE}	V_{CE}=-1V,I_C=-50mA	120	350	
		V_{CE}=-1V,I_C=-500mA	50		
Collector-emitter saturation voltage	$V_{CE(sat)}$	I_C=-500 mA, I_B= -50mA		-0.6	V
Base-emitter saturation voltage	$V_{BE(sat)}$	I_C=-500 mA, I_B= -50mA		-1.2	V
Transition frequency	f_T	V_{CE}=-6V, I_C= -20mA f=30MHz	150		MHz

CLASSIFICATION OF $h_{FE(1)}$

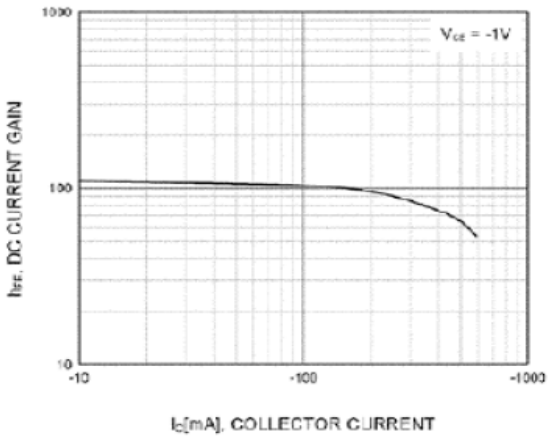
Rank	L	H
Range	120-200	200-350

Silicon Epitaxial Planar Transistor **S8550**

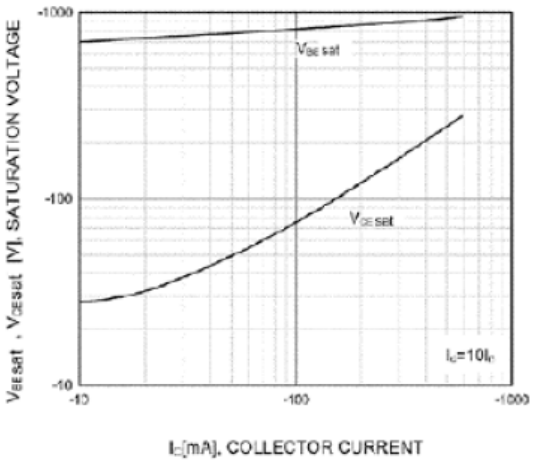
TYPICAL CHARACTERISTICS @ Ta=25℃ unless otherwise specified



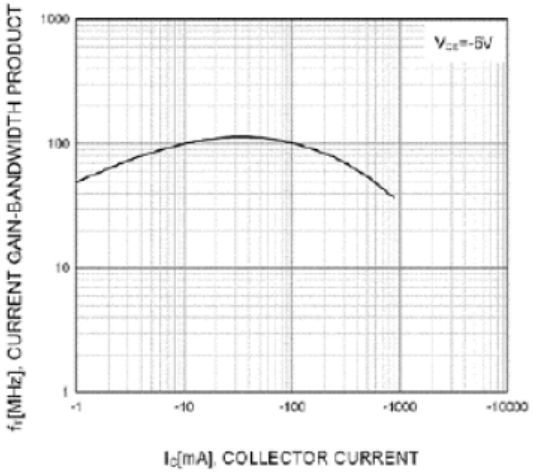
Static Characteristic



DC current Gain



Base-Emitter Saturation Voltage
Collector-Emitter Saturation Voltage

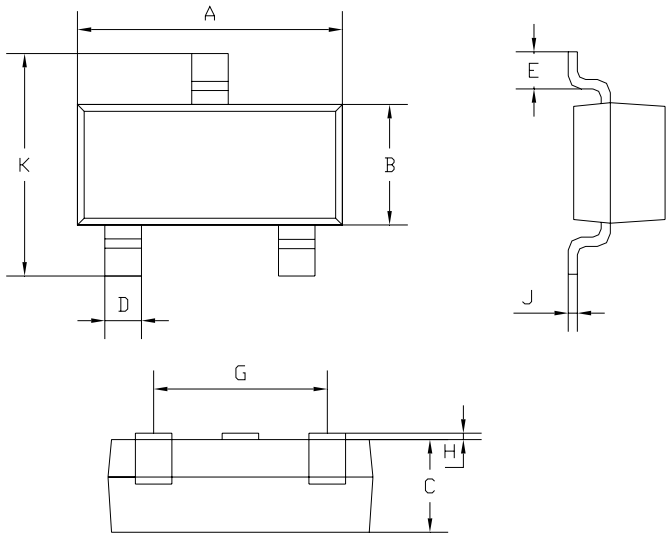


Current Gain Bandwidth Product

Silicon Epitaxial Planar Transistor **S8550**

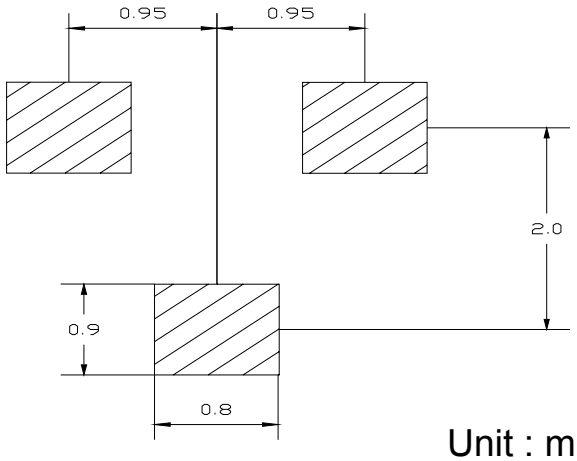
PACKAGE OUTLINE

Plastic surface mounted package SOT-23



SOT-23		
Dim	Min	Max
A	2.85	2.95
B	1.25	1.35
C	1.0Typical	
D	0.37	0.43
E	0.35	0.48
G	1.85	1.95
H	0.02	0.1
J	0.1Typical	
K	2.35	2.45
All Dimensions in mm		

SOLDERING FOOTPRINT



Unit : mm

PACKAGE INFORMATION

Device	Package	Shipping
S8550	SOT-23	3000/Tape&Reel